COMP311 Liux Lab

BIRZEIT UNIVERSITY

Project No. 2

Comp311 (Summer 2026): Linux OS Laboratory

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1 | Second Project Objectives

The following are the objectives of the first project:

- Demonstrate your understanding of Shell configuration.
- Demonstrate your ability of job control and processes.
- Demonstrate your ability to manipulate text and use regular expressions.

2 | Project Guidelines

On your virtual machine, do the following:

- Create a sub-directory under your home directory called youruserid_proj1 (e.g. **u11342145_proj2**), ***Please stick closely with the naming conventions!!***.
- Every created file must contain your full name and registration ID on the top of the first page. ***(ALL files)***.
- The file structure of what shall be delivered.

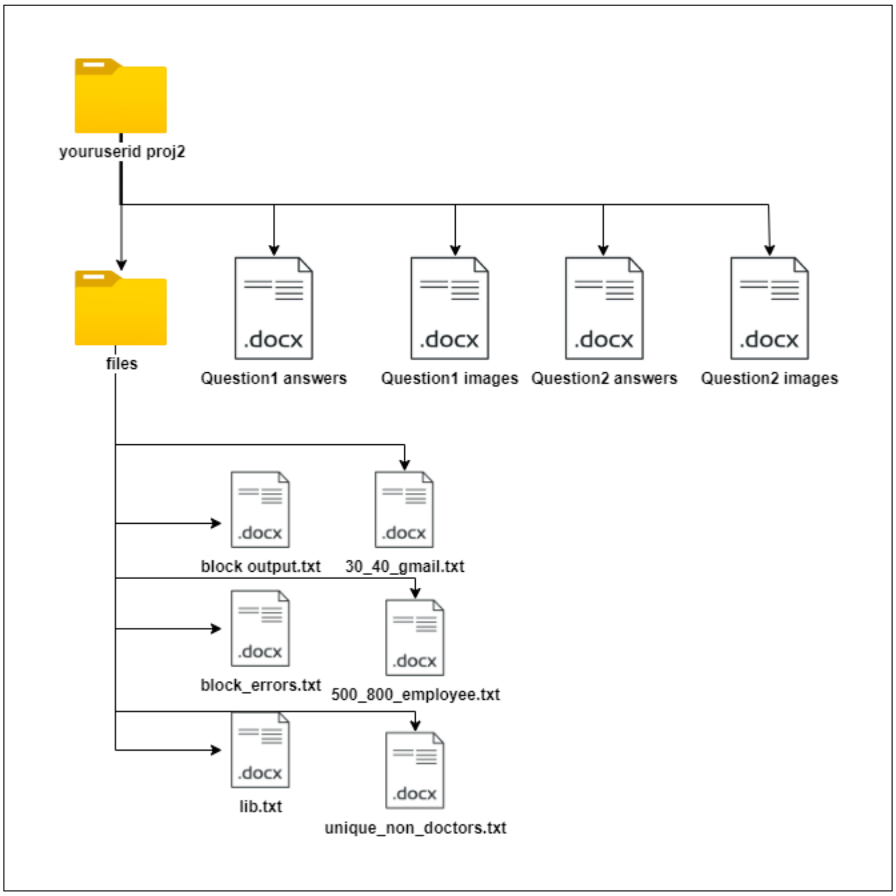

Figure 2.1: Deliverable File Structure

3 | Question 1

On your Virtual machine, document the commands necessary to perform each of the tasks outlined below. Record the number of questions and their corresponding answers in a file named **"Question1_answers"**. Additionally, ensure that you capture screenshots or images that clearly display the executed commands and the resulting outputs for each question. Organize these screenshots in a Word file named **"Question1_images"** while maintaining their sequential order.

3.1 | Part One

Q1: Using variables, find out what your hostname, username, shell, and home directories are currently set to.

Q2: Create an alias called mygroup that displays the contents of the /etc/group file on your screen in such a way that it is available every time you log in or open a new shell from your user account.

Q3:What is an initialization file? Name the two types of initialization files in Linux.

Q4:What is the difference between Job and Process in Linux OS.

Q5: Each process in the Linux OS has a Parent process, discuss what is this process called and what is the goal of this process?

3.2 | Part Two

Q1: Using the find command, search for files of type character device under the /dev directory. The command must redirect the output so that the list of character device files ends up in a file called char_output.txt.

Q2: Using the find command, search for all directories whose names start with "conf" on the entire system. The command must redirect the output so that the list of directories ends up in a file called conf.txt and the list of error messages in a file called conf_errors.txt. Provide the command you would use

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Q3: How can you use the find command to locate all files with the ".py" extension within the entire system? Provide the command to achieve this

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Q4: Using the find command, how can you identify symbolic links that end with the letter "d" within the directory /usr/lib? Provide the command to achieve this.

4 | Question 2

Given the **employees.txt** text file, use your Virtual machine to write a single command to do each of the following (Write the number of the question and the answer to a file called **(Question2_answers)**. You MUST also take screen captures or images clearly showing the commands you executed on your system for each of the questions below and whatever results were displayed on the screen. The screen captures, or images should be included in sequence in a Word file called **(Question2_images)**. Note: the columns in the employees.txt files are as follows: id,firstname,lastname,email,email2,profession

Q1: Display the full names of all users in uppercase whose first name is Reza (all cases). The output should appear like the following example REZA_LASTNAME

Q2: Display the last names in lowercase of all unique employees (all cases) with engineer as their profession, sorted by the numerical value of their ID numbers in descending order.

Q3: Display employees from row number 10 to row number 30 into an 10_30_employee.txt file. The name in this file should appear in lowercase and sorted in ascending order (i.e., first name and last name separated by a space).

Q4: Display the unique first names of all employees (all cases) whose last names end with the letter between k-w and their profession is not a doctor, and save the result to the unique_non_doctors.txt file

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Q5: Extract the email addresses of individuals whose ages are between 30 and 40 and the email domain is gmail ordered descending using their ID's. store the result in a text file called 30_40_gmail.txt

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Q6: List the first names of individuals whose last names start with "L" or "M."

Q7: Extract the professions of individuals with ages less than 50 and their profession is a firefighter

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Q8: Extract the first names of individuals with ages less than 30 and professions not containing "developer. And their first names start with "A-N"

Q9: Display the IDs of individuals with ages greater than 40 and less than 50, and whose last names contain "e"

Q10: Find the total count of individuals with ages ending in 6 or 7

Q11: Replace "officer" in professions with "cop", for all individuals whose ages end in 5

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5 | Deliverable

The project submission deadline is Saturday, September 5, 2026. All project submissions are expected to be completed using the ITC

Please turn in the file **youruserid_proj2.tar** and upload it to the ITC by the due date and time.

Note:

- You should do all the work above completely on your own.
- No projects will be accepted after the due date (05/09/2026).
- *We want to emphasize that cheating in any form is strictly prohibited. If any student is found to have engaged in cheating, whether from external sources or otherwise, appropriate actions will be taken according to our established policies.*

Good luck!